DEPARTMENT OF MATHEMATICS, IIT MADRAS

MA5950 — Mathematical Finance: Theory & Implementation

Instructor: Prof. Neelesh Upadhye

Lecture 21: Standard Brownian Motion

The Wiener Process · Nowhere Differentiable · Quadratic Variation

September 17, 2026

Today's Agenda

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2. Martingale properties

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3. Continuous everywhere, differentiable nowhere

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4. Quadratic variation and $(\mathrm{d}W)^2 = \mathrm{d}t$

Learning Objectives

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- Show W_t and $W_t^2 - t$ are martingales.
- Explain why paths are continuous but nowhere differentiable.
- Compute quadratic variation and state $(\mathrm{d}W)^2 = \mathrm{d}t$.

Standard Brownian Motion (Wiener Process)

Definition

$\{W_t\}_{t\geq 0}$ is a standard Brownian motion if:

1. $W_0 = 0$.

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2. **Independent increments** over disjoint intervals.
3. **Stationary normal increments**: $W_t - W_s \sim \mathcal{N}(0, t - s)$.
4. **Continuous paths** (with probability 1).

Each Property Earns Its Place

Remark

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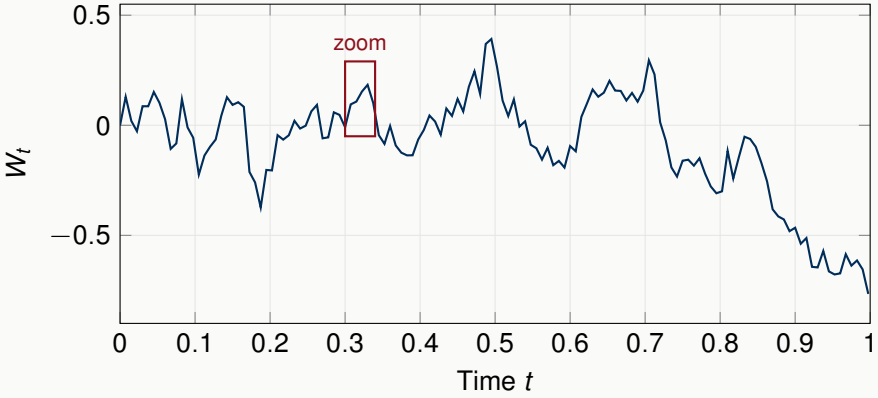
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Each Property Earns Its Place

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Property 2: “no memory.” Property 3 does the real work – increment variance $=$ elapsed time, the $\sqrt{\text{time}}$ scaling from Lecture 20. Property 4 rules out jumps. Notice what is *not* assumed: **smoothness**. That omission is a necessity, not an oversight.

A Brownian Path



Two Martingales, Again

Key Takeaway

For $s < t$:

$$\mathbb{E}[W_t \mid \mathcal{F}_s] = W_s$$

$$\mathbb{E}[W_t^2 - t \mid \mathcal{F}_s] = W_s^2 - s$$

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Write $W_t = W_s + (W_t - W_s)$, increment independent of $\mathcal{F}_s$, mean 0, variance $t - s$:

$$\mathbb{E}[W_t \mid \mathcal{F}_s] = W_s \checkmark$$

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$$\mathbb{E}[W_t^2 \mid \mathcal{F}_s] = W_s^2 + (t - s) \Rightarrow \mathbb{E}[W_t^2 - t \mid \mathcal{F}_s] = W_s^2 - s \checkmark$$

Exactly Lecture 19, in the Limit

Remark

Random walk: S_n and $S_n^2 - n$ were martingales.

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200,000 simulated paths, s=0.4, t=1, conditioning on $W_s \approx 0.5$

$$\mathbb{E}[W_t \mid W_s{=}0.5] = 0.503 \quad (\text{theory } 0.5)$$

$$\mathbb{E}[W_t^2 - t \mid W_s{=}0.5] = -0.141 \quad (\text{theory } -0.15)$$

Continuous Everywhere, Differentiable Nowhere

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The difference quotient's numerator has sd $\sqrt{h}$:

$$\frac{W_{t+h} - W_t}{h} \approx \frac{\sqrt{h}}{h} = \frac{1}{\sqrt{h}} \xrightarrow[h \to 0]{} \infty$$

Watching It Diverge

Simulated $\mathbb{E}|\Delta W/h|$ over 50,000 increments

h	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}

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Each tenfold decrease in h multiplies the quotient by $\sqrt{10}$. No limit exists.

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Structure at Every Scale

Ask for the slope of a coastline at a rock.

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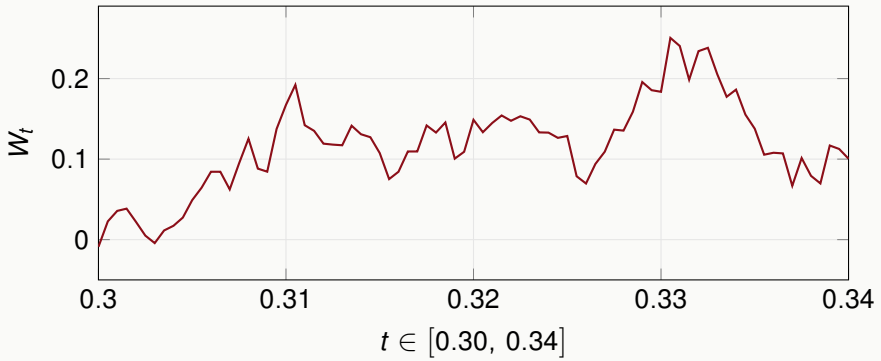
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The Same Path, Magnified $25\times$



Just as rough as the full path. No smoothing at all – hence no tangent line.

Self-Similarity

Key Takeaway

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Brownian motion is a fractal.

Quadratic Variation

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The Central Result

$$\langle W \rangle_T = T$$

while the **total variation** $\sum_i |W_{t_i} - W_{t_{i-1}}|$ **diverges**.

Both Facts, One Simulated Path

Over $[0, 1]$, refining the partition

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Squares settle to $T = 1$. Absolute values grow without bound.

Why Squaring Rescues What Absolute Value Destroys

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Each increment $\approx \sqrt{\Delta t}$; there are $n = T/\Delta t$ of them.

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Squaring is the *unique* power that neither explodes nor vanishes.

The Rule That Powers Week 8

Key Takeaway

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Why this changes everything

In ordinary calculus $(\mathrm{d}x)^2$ is negligible and gets discarded. Here it is **not** negligible – it equals $\mathrm{d}t$, a first-order quantity. Keeping this term is the entire difference between ordinary and Itô calculus.

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- **Continuous everywhere, differentiable nowhere**: quotient grows like $1/\sqrt{h}$.
- **Self-similar**: $W_{ct} \overset{d}{=} \sqrt{c}\, W_t$ – a fractal.
- $\langle W \rangle_T = T$, i.e. $(\mathrm{d}W)^2 = \mathrm{d}t$.

Check Your Understanding

Discuss (2 minutes)

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1. Compute $\mathbb{P}(W_4 > 3)$.
2. Find $\text{Cov}(W_2, W_5)$.
3. Why does $\sum(\Delta W)^2$ converge while $\sum|\Delta W|$ diverges?

Looking Ahead

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Whose surprising extra term comes straight from $(\mathrm{d}W)^2 = \mathrm{d}t$.